Astrophysics

25 questions: 8 multiple choice, 17 structured.

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1 · Part 1 Paper 1 & 2 Units [1]

Which quantity is measured in newtons (N)?

A mass

B gravitational field strength

C weight

D orbital time period

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2 · Part 1 Paper 1 & 2 Motion in the universe [1]

Which statement about the universe is correct?

A It is a large collection of billions of galaxies.

B It is a large collection of about one hundred galaxies.

C It contains only one galaxy.

D It is a large collection of planets but contains no stars.

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3 · Part 1 Paper 1 & 2 Motion in the universe [1]

Which row correctly states what each object orbits?

	a moon orbits	an artificial satellite orbits	a comet orbits
A	the Sun	the Earth	the Sun
B	a planet	the Earth	the Sun
C	a planet	the Moon	a planet
D	the Sun	the Moon	a planet

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4 · Part 1

Paper 1 & 2 | Stellar evolution | [1]

Four stars have different colours.

Which star has the highest surface temperature?

A. a red star

B. a yellow star

C. a white star

D. a blue star

5 · Part 1

Paper 2 only | Stellar evolution | [1]

The table shows the absolute magnitude of four stars.

star	absolute magnitude
P	+4.8
Q	−1.5
R	+10.2
S	+0.6

Which star gives out the most light?

A. star P

B. star Q

C. star R

D. star S

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6 · Part 1 Paper 2 only Cosmology [1]

The table shows the red-shift, Δλ / λ0, of light received from four galaxies.

galaxy	Δλ / λ0
J	0.031
K	0.012
L	0.004
M	0.019

Which galaxy is furthest from the Earth?

A galaxy J

B galaxy K

C galaxy L

D galaxy M

7 · Part 1 Paper 2 only Cosmology [1]

One star is moving towards the Earth. Its light is compared with light from the same kind of source that is not moving.

Which row describes the wavelength and the frequency of the light observed from the moving star?

	observed wavelength	observed frequency
A	longer	lower
B	longer	higher
C	shorter	lower
D	shorter	higher

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8 · Part 1

Paper 2 only

Cosmology

[1]

Astronomers use the equation

change in wavelength / reference wavelength = velocity of a galaxy / speed of light

For light from one galaxy, change in wavelength / reference wavelength = 0.020.

The speed of light is 3.0×10^{8} m/s.

What is the velocity of the galaxy?

A 6.7×10^{-11} m/s

B 6.0×10^{7} m/s

C 6.0×10^{6} m/s

D 1.5×10^{10} m/s

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9 · Part 1

Paper 1 & 2 | Motion in the universe | [7]

The table shows the gravitational field strength at the surface of four bodies in the solar system.

body	gravitational field strength in N/kg
Earth	9.8
Moon	1.6
Mars	3.7
Jupiter	25

Scientists send a rover of mass 120 kg to Mars.

(a) Calculate the weight of the rover on the surface of Mars. [3]

(b) An identical rover is sent to the Moon. State the mass of this rover on the Moon. Give a reason for your answer. [2]

(c) Explain why the gravitational field strength at the surface of Jupiter is greater than the gravitational field strength at the surface of the Earth. [2]

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10 · Part 1

Paper 1 & 2 | Motion in the universe | [7]

Planets and comets both orbit the Sun.

(a) Describe two differences between the orbit of a comet and the orbit of a planet. [2]

(b) Halley's Comet takes about 76 years to orbit the Sun once. The Earth takes 1 year.

Suggest why the time period of Halley's Comet is much longer than the time period of the Earth. [2]

(c) Explain why the speed of a comet changes as it moves around its orbit. [3]

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11 · Part 1

Paper 1 & 2 Motion in the universe [6]

The International Space Station (ISS) orbits the Earth. Its orbital radius is 6.78×10^6 m and its orbital time period is 92.5 minutes.

orbital speed = (2 × π × orbital radius) / time period

(a) Calculate the time period of the ISS in seconds. [1]

(b) Calculate the orbital speed of the ISS. [3]

(c) At the orbital radius of the ISS, the gravitational field strength is about 8.7 N/kg. At the surface of the Earth it is 9.8 N/kg.

Explain why the gravitational field strength is smaller at the orbital radius of the ISS. [2]

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12 · Part 1 Paper 1 & 2 Motion in the universe [8]

The Earth orbits the Sun with an orbital radius of 1.5×10^{11} m. Its orbital time period is 365 days.

orbital speed = (2 × π × orbital radius) / time period

(a) Show that the orbital time period of the Earth is about 3.2×10^{7} s. [2]

(b) Calculate the orbital speed of the Earth. [3]

(c) Mars orbits the Sun with an orbital radius of 2.3×10^{11} m and an orbital speed of 2.4×10^{4} m/s.

Calculate the orbital time period of Mars in days. [3]

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13 · Part 1

Paper 1 & 2 Motion in the universe [7]

This question is about a communications satellite. The satellite orbits the Earth once every 24 hours. Its orbital speed is 3.07×10^3 m/s.

orbital speed = (2 × π × orbital radius) / time period

(a) Calculate the orbital radius of the satellite. [4]

(b) The radius of the Earth is 6.4×10^6 m.

Calculate the height of the satellite above the surface of the Earth. [1]

(c) Explain what keeps the satellite moving in its orbit around the Earth. [2]

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14 · Part 1

Paper 1 & 2 Motion in the universe [4]

Our solar system is part of a galaxy.

(a) State the name of the galaxy that contains our solar system. [1]

(b) State what is meant by a galaxy. [2]

(c) The list shows four objects.

galaxy, planet, universe, solar system

State these four objects in order of size, starting with the smallest. [1]

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15 · Part 1 Paper 1 & 2 Stellar evolution [5]

The Sun is a main-sequence star. It formed from a nebula.

(a) Describe how a star such as the Sun forms from a nebula. [3]

(b) Explain why a main-sequence star such as the Sun stays the same size for a very long time. [2]

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16 · Part 1

Paper 1 & 2 Stellar evolution [5]

The Sun will reach the end of its main-sequence stage in about 5 billion years.

(a) Describe what will happen to the Sun after its main-sequence stage ends. [3]

(b) In the final stage you described in (a), the Sun will have a surface temperature much higher than its surface temperature today.

Predict the colour of the Sun at this stage. Give a reason for your answer. [2]

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17 · Part 1

Paper 1 & 2 Stellar evolution [5]

Astronomers classify stars according to their colour. The table shows the surface temperatures of four stars.

star	surface temperature in K
Betelgeuse	3500
Sun	5800
Vega	9600
Spica	22 000

(a) Identify the star in the table that appears red. Give a reason for your answer. [2]

(b) Describe how the colour of a star is related to its surface temperature. [2]

(c) The star Arcturus has a surface temperature of 4300 K.

Predict whether Arcturus appears redder or bluer than the Sun. [1]

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18 · Part 1

Paper 2 only

Stellar evolution

[6]

The table gives information about two stars, Sirius and Deneb.

star	distance from the Earth in light-years	absolute magnitude
Sirius	8.6	+1.4
Deneb	2600	−8.4

Seen from the Earth, Sirius appears much brighter than Deneb.

(a) State what is meant by absolute magnitude. [2]

(b) Identify which star gives out more light. Use data from the table to give a reason for your answer. [2]

(c) Explain why Sirius appears brighter than Deneb when seen from the Earth. [2]

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19 · Part 1 Paper 1 & 2 Stellar evolution [6]

Star Y is a main-sequence star. Its mass is about 20 times the mass of the Sun.

(a) Describe the evolution of star Y from the end of its main-sequence stage to the end of its life. [4]

(b) Explain why the end of the life of star Y is different from the end of the life of the Sun. [2]

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20 · Part 1

Paper 2 only

Cosmology

[5]

The siren of an ambulance emits sound of frequency 700 Hz. The ambulance moves away from an observer at a constant speed. The observer is standing at the side of the road.

(a) Describe how the frequency and the wavelength of the sound heard by the observer compare with the frequency and the wavelength of the sound emitted by the siren. [2]

(b) Explain why the wavelength of the sound reaching the observer is different from the wavelength of the sound emitted. [2]

(c) The driver of the ambulance hears the siren at a frequency of 700 Hz.

Explain why. [1]

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21 · Part 1

Paper 2 only

Cosmology

[6]

In a laboratory, a line in the spectrum of hydrogen has a wavelength of 656 nm. In light received from galaxy G, the same line has a wavelength of 669 nm.

change in wavelength / reference wavelength = velocity of a galaxy / speed of light

The speed of light is 3.0×10^8 m/s.

(a) Calculate the change in wavelength. [1]

(b) Calculate the velocity of galaxy G. [3]

(c) State whether galaxy G is moving towards the Earth or away from the Earth. Give a reason for your answer. [2]

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22 · Part 1

Paper 2 only

Cosmology

[6]

Galaxy H is moving away from the Earth at a velocity of 2.4×10^7 m/s.

In a laboratory, one line in the spectrum of hydrogen has a wavelength of 486 nm.

change in wavelength / reference wavelength = velocity of a galaxy / speed of light

The speed of light is 3.0×10^8 m/s.

(a) Calculate the wavelength of this line in the light received from galaxy H. [4]

(b) Another line in the spectrum of hydrogen has a wavelength of 434 nm in the laboratory.

Calculate the wavelength of this line in the light received from galaxy H. [2]

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23 · Part 1

Paper 2 only

Cosmology

[6]

The table shows the distance from the Earth of four galaxies and the red-shift, $\Delta\lambda / \lambda 0$, of the light received from each galaxy.

galaxy	distance from the Earth in millions of light-years	$\Delta\lambda / \lambda 0$
1	120	0.009
2	250	0.019
3	390	0.029
4	510	0.038

(a) Describe the relationship between the distance of a galaxy from the Earth and the red-shift of its light. Use data from the table in your answer. [2]

(b) Explain why the red-shift of galaxies provides evidence that the universe is expanding. [4]

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24 · Part 1 Paper 2 only Cosmology [5]

Scientists detect cosmic microwave background (CMB) radiation.

(a) Describe the CMB radiation. [2]

(b) Explain how the Big Bang theory accounts for the CMB radiation. [3]

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25 · Part 1 Paper 2 only Cosmology [5]

The Big Bang theory describes the past evolution of the universe.

(a) Describe how the universe has changed from the Big Bang to the present day. [3]

(b) The universe is still expanding.

Predict how the wavelength of the CMB radiation will change in the future. Give a reason for your answer. [2]

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